

GEORGE RUSU

AI Engineer — Agentic Automation • LangGraph, RAG & HITL in Production

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PROFESSIONAL SUMMARY

AI Engineer who builds autonomous agents for real operations, not demos. Four years of enterprise automation with UiPath, n8n and Python, removing up to 85% of manual workload and cutting time and costs by up to 80%, followed by production agentic AI: LangGraph pipelines, LLM decision layers and Human-in-the-Loop approvals running inside live business operations. The value is the combination in one engineer: I speak the language of operations (processes, exceptions, audits, SLAs) and the language of agents (orchestration, retrieval, tool calling, observability), so the systems I ship are reliable enough for the business to actually depend on. Available immediately for AI engineering and agentic automation roles, remote or hybrid.

IMPACT SNAPSHOT

- **85%** — manual workload removed from enterprise business processes through UiPath and n8n automation (ATOS).
- **80%** — time and cost saved via a continuous automation-improvement pipeline across global teams.
- **Agentic AI in production** — LangGraph pipelines with Human-in-the-Loop approvals shipped into live business operations; public work documented with live demos on my portfolio.
- **Top 3% on TryHackMe** — reached in 3 months; security thinking applied to agent permissions, secrets and failure modes.

WORK EXPERIENCE

AI Integration Engineer — B2B operations platform for manufacturers

June

2026 – August 2026

SaaS platform automating manufacturer operations: sales, logistics and raw-material replenishment through a single API.

- **End-to-end agent workflows** — designed and shipped AI agent pipelines automating manufacturer operations across the unified API platform, replacing manual coordination between sales, logistics and procurement.
- **Predictive replenishment** — built agentic pipelines with LangGraph and FastAPI forecasting raw-material needs and triggering replenishment before stock-outs, turning reactive ordering into forecast-driven decisions.
- **LLM decision layers** — embedded reasoning steps into logistics and sales flows so routine operational judgement calls were resolved autonomously, with every decision traceable.
- **Human-in-the-Loop by design** — architected approval systems for critical operational decisions: agents acted autonomously where risk was low and escalated to a human where it was not.
- **Production reliability** — instrumented agent runs with tracing and metrics; defined retries, fallbacks and idempotency so failures degraded safely instead of corrupting operational data.

Stack: LangGraph • FastAPI • Python • LLM APIs • RAG • Redis • Docker • n8n • Langfuse • REST integrations

RPA Developer, AI Automation Team — ATOS

May

2022 – June 2026

Enterprise-grade automation at a global technology corporation, delivered for and with distributed international teams.

- **UiPath** — built, deployed and maintained enterprise robots across live business processes, cutting manual workload by up to 85%.
- **n8n** — orchestrated the integration layer: APIs, queues and internal systems connected into end-to-end automated flows between the robots and the business.
- **Python** — used where RPA hit its limits: data cleaning and transformation, parsing, validation logic and API calls on large business-critical datasets.
- **Continuous improvement** — ran an ongoing optimisation pipeline on production automations, delivering up to 80% savings in processing time and operational cost.
- **Reliability and handover** — designed automation architectures that absorbed growing demand without added headcount, documented in technical specs and SOPs used by distributed global teams.
- **RPA to agentic AI** — drove the shift from rule-based automation to LLM-driven systems, carrying enterprise standards (monitoring, error handling, auditability) into agent design.

Stack: UiPath • n8n • Python • REST APIs • Azure • data processing • SOP and technical documentation

Earlier career — Chef / Team Coordinator (Mummelsee, Germany; Reya Event Club)

2012

– 2020

- Eight years leading teams of up to 20 people in high-pressure, high-output operations: 100+ covers a day, zero tolerance for process failure. This is where the operational discipline, multicultural communication and rapid decision-making behind the engineering work were built.

SELECTED AI PROJECTS (2026)

Agent Orchestrator — Multi-Agent System [Live] | LangGraph · Groq · FastAPI · ChromaDB · SSE · HF Spaces

- LangGraph orchestrator routes each visitor request to the right specialist agent: RAG Q&A over my background, Gmail agent that sends the CV, Calendar agent that books a Google Meet, Job-Match agent that scores fit against a job description, Telegram agent for instant notifications.
- Human-in-the-Loop calendar approval through a Telegram bot with Upstash Redis state; every routing decision streams live to the UI over SSE.
- Value: a static portfolio turned into a 24/7 agent that answers recruiters, scores the role and books the meeting without me in the loop.

Production RAG Service — Hybrid Search API [Shipped] | FastAPI · Qdrant · Redis · Langfuse · github.com/georgelush/rag-system

- Multi-tenant RAG API built from first principles with the full 2026 stack: hybrid dense + sparse retrieval with RRF fusion, cross-encoder reranking sidecar and streaming SSE answers.
- Async ingest queue with dead-letter handling, namespace-scoped RBAC with JWT, Prometheus metrics and per-stage Langfuse tracing; 13 file formats including audio via Whisper.
- Value: retrieval quality and tenancy isolation that a company can put in front of customers, not a notebook demo.

AI On-Call — Voice Agent for IT Support [Pilot case] | Twilio · LangGraph · n8n · ITSM · Redis

- Outbound voice agent that fires the moment an ITSM ticket opens: calls the customer, verifies identity with a 6-digit code, collects the work note and ETA, then writes everything back into ITSM through n8n.
- Value: removes the human dispatcher from first contact and shortens response time on every ticket.

Agentic AI Playground [Shipped] | LangGraph · LangChain · n8n · Qdrant · Langfuse · Gradio

- Working lab for agentic patterns: orchestration graphs, hybrid retrieval on Qdrant and end-to-end Langfuse observability, used to validate patterns before they reach production.

TECHNICAL SKILLS

Agentic AI: LangGraph, LangChain, multi-agent orchestration, Human-in-the-Loop design, tool and function calling, autonomous workflow design, agent evaluation

LLM & RAG: RAG pipelines, hybrid dense+sparse search (RRF), cross-encoder reranking, vector databases, embedding models, prompt engineering, Langfuse tracing and observability, OpenAI API, Groq API, Whisper

Backend & Infrastructure: Python, FastAPI, REST APIs, SSE streaming, Docker and Docker Compose, CI/CD, Prometheus, Azure, Hugging Face Spaces

Data & Storage: Qdrant, ChromaDB, Redis, Upstash, data engineering, large-scale data cleaning and transformation

Automation & Integrations: UiPath (enterprise RPA), n8n, process automation, SOP design, Twilio, Gmail API, Google Calendar API, Telegram Bot API, FastMCP, ITSM

Security & Systems: Networking fundamentals, Linux administration, web exploitation and privilege escalation labs (TryHackMe, top 3%), C++ and Java fundamentals, OOP, algorithms

Ways of working: AI-assisted development (Claude CLI, VS Code, GitHub), technical documentation, English (professional working proficiency), Romanian (native)

EDUCATION & CERTIFICATIONS

- **AI Engineer for Developers** — DataCamp certification, March 2026 (ID AIEDA0019097326736).
- **WellCode Programming Mentorship** — instructors from Meta, Google and Palantir; C++ and Java fundamentals, OOP, algorithms, 200+ coding challenges solved, Unreal Engine 5 prototypes.
- **TryHackMe** — top 3% globally within 3 months (2025–2026); networking, Linux, web exploitation, privilege escalation, fully documented lab notes.